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In [1]: import pandas as pd
students = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx", sheet_n
marks = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx", sheet_n
contact = pd.read_excel("Merge_Join_Concat_Dataset_50_Records.xlsx", sheet
#1. Inner Merge
inner_df = pd.merge(
    students,
    marks,
    on="Student_ID",
    how="inner"
)
print(inner_df.head())
print(inner_df.shape)
#2. Left Merge
left_df = pd.merge(
    students,
    marks,
    on="Student_ID",
    how="left"
)
print(left_df.tail())
print(left_df.shape)
#Right Merge
right_df = pd.merge(
    students,
    marks,
    on="Student_ID",
    how="right"
)
print(right_df.tail())
print(right_df.shape)
#4.Outer Merge
outer_df = pd.merge(
    students,
    marks,
    on="Student_ID",
    how="outer"
)
print(outer_df)
print(outer_df.shape)
#5.Merge Multiple table
student_complete = pd.merge(
    students,
    marks,
    on="Student_ID",
    how="left"
)
student_complete = pd.merge(
    student_complete,
    contact,
    on="Student_ID",
    how="left"
)
print(student_complete.head())
#Concat Row-Wise
#Split students into two parts.
part1 = students.iloc[:25]
part2 = students.iloc[25:]
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combined_rows = pd.concat([part1, part2])
print(combined_rows.shape)
# Concat Column-Wise
student_basic = students[['Student_ID', 'Name']]
student_marks = marks[['Python']]
combined_cols = pd.concat(
    [student_basic, student_marks],
    axis=1
)
print(combined_cols.head())
#join()
#join() works using indexes.
students_index = students.set_index("Student_ID")
contact_index = contact.set_index("Student_ID")
joined_df = students_index.join(contact_index)
print(joined_df.tail(10))
```

|   | Student_ID | Name      | Department | Maths | Python |
|---|------------|-----------|------------|-------|--------|
| 0 | 1001       | Student_1 | CSE        | 61    | 56     |
| 1 | 1002       | Student_2 | IT         | 62    | 57     |
| 2 | 1003       | Student_3 | ECE        | 63    | 58     |
| 3 | 1004       | Student_4 | EEE        | 64    | 59     |
| 4 | 1005       | Student_5 | MECH       | 65    | 60     |

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|    | Student_ID | Name       | Department | Maths | Python |
|----|------------|------------|------------|-------|--------|
| 45 | 1046       | Student_46 | CSE        | NaN   | NaN    |
| 46 | 1047       | Student_47 | IT         | NaN   | NaN    |
| 47 | 1048       | Student_48 | ECE        | NaN   | NaN    |
| 48 | 1049       | Student_49 | EEE        | NaN   | NaN    |
| 49 | 1050       | Student_50 | MECH       | NaN   | NaN    |

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|    | Student_ID | Name | Department | Maths | Python |
|----|------------|------|------------|-------|--------|
| 45 | 2001       | NaN  | NaN        | 75    | 80     |
| 46 | 2002       | NaN  | NaN        | 75    | 80     |
| 47 | 2003       | NaN  | NaN        | 75    | 80     |
| 48 | 2004       | NaN  | NaN        | 75    | 80     |
| 49 | 2005       | NaN  | NaN        | 75    | 80     |

(50, 5)

|    | Student_ID | Name       | Department | Maths | Python |
|----|------------|------------|------------|-------|--------|
| 0  | 1001       | Student_1  | CSE        | 61.0  | 56.0   |
| 1  | 1002       | Student_2  | IT         | 62.0  | 57.0   |
| 2  | 1003       | Student_3  | ECE        | 63.0  | 58.0   |
| 3  | 1004       | Student_4  | EEE        | 64.0  | 59.0   |
| 4  | 1005       | Student_5  | MECH       | 65.0  | 60.0   |
| 5  | 1006       | Student_6  | CSE        | 66.0  | 61.0   |
| 6  | 1007       | Student_7  | IT         | 67.0  | 62.0   |
| 7  | 1008       | Student_8  | ECE        | 68.0  | 63.0   |
| 8  | 1009       | Student_9  | EEE        | 69.0  | 64.0   |
| 9  | 1010       | Student_10 | MECH       | 70.0  | 65.0   |
| 10 | 1011       | Student_11 | CSE        | 71.0  | 66.0   |
| 11 | 1012       | Student_12 | IT         | 72.0  | 67.0   |
| 12 | 1013       | Student_13 | ECE        | 73.0  | 68.0   |
| 13 | 1014       | Student_14 | EEE        | 74.0  | 69.0   |
| 14 | 1015       | Student_15 | MECH       | 75.0  | 70.0   |
| 15 | 1016       | Student_16 | CSE        | 76.0  | 71.0   |
| 16 | 1017       | Student_17 | IT         | 77.0  | 72.0   |
| 17 | 1018       | Student_18 | ECE        | 78.0  | 73.0   |
| 18 | 1019       | Student_19 | EEE        | 79.0  | 74.0   |
| 19 | 1020       | Student_20 | MECH       | 80.0  | 75.0   |
| 20 | 1021       | Student_21 | CSE        | 81.0  | 76.0   |
| 21 | 1022       | Student_22 | IT         | 82.0  | 77.0   |
| 22 | 1023       | Student_23 | ECE        | 83.0  | 78.0   |
| 23 | 1024       | Student_24 | EEE        | 84.0  | 79.0   |
| 24 | 1025       | Student_25 | MECH       | 85.0  | 80.0   |
| 25 | 1026       | Student_26 | CSE        | 86.0  | 81.0   |
| 26 | 1027       | Student_27 | IT         | 87.0  | 82.0   |
| 27 | 1028       | Student_28 | ECE        | 88.0  | 83.0   |
| 28 | 1029       | Student_29 | EEE        | 89.0  | 84.0   |
| 29 | 1030       | Student_30 | MECH       | 90.0  | 85.0   |
| 30 | 1031       | Student_31 | CSE        | 91.0  | 86.0   |
| 31 | 1032       | Student_32 | IT         | 92.0  | 87.0   |
| 32 | 1033       | Student_33 | ECE        | 93.0  | 88.0   |
| 33 | 1034       | Student_34 | EEE        | 94.0  | 89.0   |
| 34 | 1035       | Student_35 | MECH       | 95.0  | 90.0   |
| 35 | 1036       | Student_36 | CSE        | 96.0  | 91.0   |
| 36 | 1037       | Student_37 | IT         | 97.0  | 92.0   |
| 37 | 1038       | Student_38 | ECE        | 98.0  | 93.0   |

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38      1039 Student_39      EEE  99.0  94.0
39      1040 Student_40      MECH 60.0  95.0
40      1041 Student_41      CSE  61.0  96.0
41      1042 Student_42      IT   62.0  97.0
42      1043 Student_43      ECE  63.0  98.0
43      1044 Student_44      EEE  64.0  99.0
44      1045 Student_45      MECH 65.0  55.0
45      1046 Student_46      CSE   NaN   NaN
46      1047 Student_47      IT   NaN   NaN
47      1048 Student_48      ECE   NaN   NaN
48      1049 Student_49      EEE   NaN   NaN
49      1050 Student_50      MECH  NaN   NaN
50      2001      NaN      NaN  75.0  80.0
51      2002      NaN      NaN  75.0  80.0
52      2003      NaN      NaN  75.0  80.0
53      2004      NaN      NaN  75.0  80.0
54      2005      NaN      NaN  75.0  80.0

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(55, 5)
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      Student_ID      Name Department  Maths  Python      Email
0      1001 Student_1      CSE  61.0  56.0 student1@college.edu
1      1002 Student_2      IT   62.0  57.0 student2@college.edu
2      1003 Student_3      ECE  63.0  58.0 student3@college.edu
3      1004 Student_4      EEE  64.0  59.0 student4@college.edu
4      1005 Student_5      MECH 65.0  60.0 student5@college.edu

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      Student_ID      Name  Python
0      1001 Student_1      56
1      1002 Student_2      57
2      1003 Student_3      58
3      1004 Student_4      59
4      1005 Student_5      60

      Name Department      Email
Student_ID
1041      Student_41      CSE  student41@college.edu
1042      Student_42      IT   student42@college.edu
1043      Student_43      ECE  student43@college.edu
1044      Student_44      EEE  student44@college.edu
1045      Student_45      MECH student45@college.edu
1046      Student_46      CSE  student46@college.edu
1047      Student_47      IT   student47@college.edu
1048      Student_48      ECE  student48@college.edu
1049      Student_49      EEE  student49@college.edu
1050      Student_50      MECH student50@college.edu

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In [ ]:
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